

BAO NGUYEN

tbnguye1@ualberta.ca | linkedin.com/in/bao-thanh-nguyen | github.com/BaoTNguyen

Relevant Experience

M2M Tech

Surrey, BC

Machine Learning Engineer Intern

Apr 2024 - Present

- Developed an end-to-end integration of digital twin models using **Azure** (IoT Central, Digital Twins, Machine Learning, Data Explorer) and Power BI to integrate real-time data analysis and optimization using machine learning
- Compiled and trained a dataset of 100 articles using **LangChain** and self-critique prompting to extract domain-specific entity information, then configured and presented results using **LangSmith**
- Implemented a data pipeline with **LLM classification** for 790 prompts to extract high-quality prompts for **web scraping**, display Google and Bing scraping results, and determine prompt quality using **ROUGE** and other metrics
- Transitioned code into production for handover by organizing an existing codebase into the desired code architecture, providing technical and functional documentation, as well as communicated effectively to satisfy project requirements

NXT-GEN, University of Alberta

Edmonton, AB

Vice President Marketing

June 2022 - Aug 2024

- Led a marketing team of 5 students to develop and execute marketing campaigns, ensuring timely progress and communication through weekly meetings
- Reached out to 20 high schools and recruited over 50 students into the Accelerator Program, an entrepreneurship initiative with over \$6,000 in competition prizes for high school and university students
- Advertised events and competitions as well as maintained the organization's social media presence

Wayfinders Business Co-operative

Edmonton, AB

Project Budget Planner Intern

Dec 2023 - Feb 2024

- Prepared **Excel** financial budgets for entrepreneurial events and projects, satisfying all internal requirements and reducing costs by 20%
- Revised the company's general budget, resulting in a 10% cost decrease after updating current revenue streams and optimizing expenses
- Collected a list of grants and loans worth over \$100,000 to help raise funds for the above projects

Youreka Canada

Edmonton, AB

Undergraduate Research Lead

Feb 2023 - Apr 2023

- Wrote a 10-page research project with 3 high school students on mental health services for Canadian youth over the COVID-19 pandemic
- Analyzed government data using **R** libraries and presented findings at a research symposium with over 50 attendees

Education

University of Alberta

Edmonton, AB

Bachelors of Science Specialization in Mathematics and Finance

Expected Graduation: Jun 2026

Microsoft Certified: Azure AI Fundamentals (AI-900)

Jan 2024

- Certification by Microsoft for foundational knowledge in AI and machine learning concepts using Azure services

Personal Projects

Canadian Rental Price Prediction

Oct 2023 - Dec 2023

- Created a mathematical model based on the Markov voter model to predict average rental prices of cities in Canada
- Performed maximum likelihood estimation in **Python**

Standard Metropolitan Statistical Area

Oct 2023 - Dec 2023

- Predicted serious crime rate using a linear regression model with 11 independent variables in **R**
- Compiled a 15-page report about the case study, methodology, and results

Technical Skills

Languages: Python, R, SQL

Technologies: Microsoft Azure, Excel, Power BI, Linux, Git, VS Code, NumPy, Pandas, TensorFlow, Scikit-learn, LangChain